

A Quantum Allosteric Scanner for Undruggable Targets

Continuous-time quantum walks with learned operator selection

Team AuraQu | Cleveland Clinic problem statement: *Unlocking undruggable targets: quantum simulation of allosteric signal propagation*

1. Problem Framing

A cryptic allosteric pocket is useful precisely when it is *distal*: far enough from the active site that no geometric pocket-finder looking near the catalytic machinery will locate it. This is not a rhetorical point, and we measured it. Ranking residues purely by graph distance from the active site — the strongest “free” classical heuristic — achieves **AUC 0.227** on distal targets across 53 protein families. That is not merely weak; it is *below chance*, because a distal pocket is by definition where proximity is not. Geometry does not underperform on this problem, it points the wrong way.

That is the gap a quantum method can fill, and it is why the challenge’s own success definition names *distal* regulatory residues specifically. Signal propagation through a protein contact network is a transport problem on a graph. A continuous-time quantum walk (CTQW) explores such a graph by amplitude interference rather than diffusion: paths cancel and reinforce, so the walker distinguishes routes that a classical random walk — which only accumulates probability — cannot. Our hypothesis is that this interference structure is what carries allosteric information, and our evidence is that the walk succeeds exactly where the distance heuristic inverts.

2. Technical Approach

Paradigm. Hybrid quantum–AI. The physics layer is a CTQW, currently executed by exact spectral simulation and targeted for Trotterised circuit execution in Phase 2 (§3). The learning layer is a classical regressor that selects which quantum operator to run. We state plainly that today’s results come from simulation, not hardware; §3 gives the qubit and depth budget and the path to execution.

The walk. From the apo structure alone we build a $C\alpha$ contact graph at 8 Å (elastic-network assumption, as the challenge specifies) and a Hamiltonian H from its adjacency or Laplacian, optionally with a diagonal site potential V encoding B-factor disorder, solvent exposure, rigidity and low-mode participation. Seeding at the active site, the walker evolves as $U(t) = e^{-iHt}$ and we score every candidate residue by its dynamic connectivity to that site.

A single evolution time is unsafe: destructive interference can drive a genuine coupling to zero, and on disordered graphs the walk can Anderson-localise. We therefore use time-averaged quantities, including the closed form $p_{\text{avg}}(s \rightarrow a) = \sum_k |v_k(s)|^2 |v_k(a)|^2$, which is parameter-free and provably free of cancellation. Alongside it we score Green’s functions $|G(E, \eta)|^2$, walker dispersion diagnostics, and a spectral variance $\Delta E = \sqrt{\langle E^2 \rangle - \langle E \rangle^2}$ that proved to be our single strongest

score.

The operator is not universal — and that is the central technical finding. We evaluated 13 Hamiltonians $\times$ 17 scores $\times$ 4 seed-filter settings = 884 configurations per protein on 630 proteins. A variance decomposition is decisive: main effects (“this Hamiltonian is generally good”) explain **1%** of the variance in performance; the remaining **99%** is interaction and per-protein variation. Across 630 proteins there are **308 distinct winning configurations**. There is no single right operator to publish.

The AI layer follows directly from that. Rather than fix an operator or search all 884, we train a model to *recommend* configurations from protein topology alone. Input: nine graph features (size, edges, degree statistics, clustering, diameter, algebraic connectivity, spectral radius). Output: a ranked, weighted shortlist of k configurations to run, k chosen by the user. A feasibility gate first removes settings the protein cannot support — on a compact protein spanning five hops, an aggressive seed filter deletes the answer before the walk starts.

3. Feasibility and Resource Requirements

Already demonstrated. The pipeline described above has been run end to end on a 1022-protein worklist; 630 proteins across 399 families produce a scoreable result. Total cloud compute for every experiment reported here is under **EUR 3**. The classical simulation cost is dominated by one eigendecomposition per operator per protein, which is seconds at these sizes. Feasibility of the *classical* half is therefore not a projection — it is measured.

The quantum half, honestly. Mapping one residue to one qubit is not viable for a 500-residue target on near-term hardware. Our Phase 2 plan is basis encoding of the seed with Trotterised evolution, $e^{-iHt} \approx (\prod_j e^{-iH_j t/r})^r$, where H decomposes over graph edges. At $r = 2$ we already measure $\sim 10^{-3}$ agreement with exact evolution in simulation, so depth stays shallow by construction. Coarse-graining is the enabling step and we report a genuine failure: a 16-node community-level walk collapsed to AUC 0.529, because scoring at community level and back-propagating to residues forces ties. Fixing this — compressing while proving the topological signal survives — is the principal Phase 2 risk and the principal Phase 2 task.

Resources needed. AWS Braket and Classiq access as provided; one modest CPU node for the classical layer. No proprietary data: every structure is public RCSB, and no classical MD trajectory enters the pipeline, per the challenge constraint.

4. Expected Impact

A successful PoC delivers a scanner that ingests an apo structure of a shelved target and returns a ranked probability map of candidate allosteric sites, upstream of high-throughput screening. The quantitative target we propose for Phase 2 follows from what we already achieve: on distal targets the walk must beat the geometric floor with family-level significance, which it currently does at $p < 10^{-8}$, and it must place true pocket residues in the top five often enough to direct library design. We are explicit that the second half is where the current method is weakest (§5), and that is the honest measure of what Phase 2 must improve.

5. Validation Plan

Our validation discipline is the part of this proposal we would most want a reviewer to examine, because the failure mode in this field is a number that survives no control.

Counting. Structures are grouped into protein families and every result is counted per family, never per structure. This is not cosmetic: one family in our set contributes 28 structures of a single protein, and per-structure counting inflated an early result by roughly twentyfold.

Controls actually run. Label-permutation nulls preserving score correlation; family-level permutation nulls; a favourite-pocket null; a random-residue-through-the-same-pipeline null; and the proximity floor described in §1. Classical baselines — proximity, degree, pocket size, fpocket druggability, PASSer rank — are scored on the identical residues and labels.

Blind evaluation. Every model number is leave-one-family-out cross-validated: each protein is scored by a model that never saw its family.

Headline results, and their limits.

We state the negative half plainly: *overall* the walk does not beat distance. The result is specifically and only about distal targets — which is what the challenge asks about, and which is the case classical geometry cannot address.

On held-out targets never seen in training, the model's six recommended configurations were run through the full pipeline. On **BCR-ABL1** (1OPL→5MO4, asciminib) the best of the six reached **AUC 0.900, P@5 0.80** — four of the top five residues are genuine myristoyl-pocket contacts. On **HIV-1 reverse transcriptase** the best reached AUC 0.679, matching an independent pre-registered run, but with no top-five hit.

What we found wrong, and report anyway. The recommender's *ranking* works; its *confidence weighting* does not — on BCR-ABL1 the winning configuration sat at rank 4 with 7% weight while the top-weighted pick scored AUC 0.140. Merging the shortlist is worse than picking one from it. And two defects in the challenge's own Table 1 warrant flagging to the organisers: **6C1H contains no mavacamten** (only ADP and MG, verified live against RCSB), and **4OBE is wild-type KRAS**, not G12C — residue 12 is glycine, so the named apo structure is the wrong genotype for the stated ob-

jective.

6. Hybrid / Cross-Domain Integration

The hybrid design is not decorative. The quantum layer produces evidence — a residue ranking per operator — and the AI layer decides which evidence to gather, reducing 884 candidate quantum computations to k (typically 6), a $147\times$ reduction. Measured leave-one-family-out, $k=6$ retains a configuration reaching $\text{AUC} \geq 0.8$ for 45% of proteins and recovers 43 of the 71 families that exhaustive search finds; $k=20$ recovers 54. This matters more, not less, on quantum hardware, where each configuration is a separate circuit and exhaustive sweeps are impossible. HPC orchestration runs the classical sweeps that generate the model's training data.

7. Team Capability

[PLACEHOLDER — team names, roles, affiliations, and prior quantum/domain experience to be supplied. Note the guidelines require a separate Team Profile section in the portal in addition to this.]

	proximity floor	CTQW (blind)	families where CTQW wins
Distal targets (91 proteins, 53 families)	0.227	0.617	47/53 , $p < 10^{-8}$
Near targets (539 proteins, 350 families)	0.633	0.597	137/350 (floor wins)
All (630 proteins, 399 families)	0.574	0.600	182/399, $p = 0.09$